

# DEBANKUR DUTTA

AI/ML Engineer | Backend & Intelligent Systems

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## PROFESSIONAL SUMMARY

AI/ML Engineer specializing in backend-driven intelligent systems with experience building production-ready AI applications using **Python**, **FastAPI**, **TensorFlow**, and modern LLM workflows. Skilled in **Retrieval-Augmented Generation (RAG)**, embeddings, vector databases, intelligent agents, and scalable API development. Passionate about building performant AI systems for real-world applications.

## EDUCATIONS

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|-------------------------------------------------------------------------------------------------------|---------------------|
| <b>JIS University</b>   Kolkata<br><i>B.Tech in Computer Science &amp; Engineering (Data Science)</i> | Aug 2023 - Jun 2027 |
| <b>Aditya Academy</b>   Kolkata<br><i>Higher Secondary</i>                                            | Apr 2022 - Apr 2023 |
| <b>Aditya Academy</b>   Kolkata<br><i>Secondary</i>                                                   | Apr 2020 - Apr 2021 |

## PROJECTS

### Travel AI Agent | 🌐 Website 🐙 Github

*Agentic Backend System*

- Developed an **agentic AI travel assistant** using **LangChain**, **FastAPI**, and LLM tool-calling to automate itinerary generation and personalized travel recommendations.
- Engineered an **asynchronous FastAPI backend** with **Redis caching**, **reducing response latency by approximately 50%** during load testing.
- Implemented persistent conversational memory with **Supabase** and dynamic LLM routing, enabling context-aware recommendations while **reducing inference costs by approximately 30%**.

*Technologies / Tools Used : LangChain, FastAPI, Supabase, MongoDB, Python, OpenAI, Llama API, Docker*

### Poneglyph – Multi-Tenant Hybrid RAG Platform | 🌐 Website 🐙 Github

- Built a production-ready **multi-tenant Hybrid RAG** platform enabling secure document ingestion, tenant-aware retrieval, and intelligent querying over enterprise knowledge bases.
- Architected a hybrid retrieval pipeline using **BM25**, **dense vectors**, **RRF**, and **reranking**, improving retrieval relevance by **~30%** over dense search alone.
- Developed an async ingestion pipeline with **chunking**, **embeddings**, **vector indexing**, and **Redis caching**, reducing query latency by **~40%**.
- Implemented **strict tenant-aware document access** using **FastAPI**, **PostgreSQL**, **Qdrant**, and **Docker**, ensuring users could retrieve **only their own organization's data**.

*Technologies / Tools Used : FastAPI, LangChain, QdrantDB, Docker, ELK Stack, Rerank, BM25, RRF*

### AI Image Detector

*Batch Inference System*

- Built an **AI image detection platform** using **EfficientNet transfer learning**, achieving **89.87% classification accuracy**.
- Engineered **asynchronous FastAPI inference APIs**, improving concurrent request throughput by **2×** for batch image processing.
- Optimized CPU-only inference with automated Resend reporting, delivering sub-second predictions without GPU infrastructure.

*Technologies / Tools Used : FastAPI, TensorFlow, Supabase, MongoDB, Resend, Docker*

## SKILLS

|                                     |                                                                      |
|-------------------------------------|----------------------------------------------------------------------|
| <b>Databases :</b>                  | MongoDB, PostgreSQL, MySQL, Supabase, Elasticsearch, QdrantDB, Redis |
| <b>Frameworks &amp; Libraries :</b> | FastAPI, Express.js, Next.js, React.js, Fastapi, LangChain           |
| <b>Languages :</b>                  | Hindi, English                                                       |
| <b>Programming Languages :</b>      | Python, JavaScript, SQL                                              |
| <b>Tools &amp; Platforms :</b>      | Git, GitHub, Postman, Docker, MLFlow                                 |